

SMART AUTOMATIC STREET LIGHT CONTROL SYSTEM USING LDR

CodeAlpha IoT Internship Project

Abstract

The Smart Automatic Street Light Control System is an Arduino-based automation project designed to reduce energy consumption by automatically controlling street lights according to ambient light conditions. The system uses an LDR (Light Dependent Resistor) to detect environmental light intensity and automatically switches street lights ON during darkness and OFF during daylight. This project was designed, implemented, and tested using the Tinkercad simulation platform as a virtual prototype and proof of concept.

Components Used

Arduino Uno, LDR (Light Dependent Resistor), LED, Resistors, Breadboard, and Jumper Wires.

Software Used

Arduino IDE, Embedded C, and Tinkercad Simulation.

Features

Automatic Street Light Control, Day and Night Detection, LDR Sensor Interfacing, Energy-Efficient Operation, Reduced Power Consumption, and Fully Automated Functionality.

Future Scope

This project has been successfully designed and tested in Tinkercad simulation. In the future, I plan to implement this project using real hardware components to validate its performance in real-world conditions and gain practical experience in hardware deployment.

Conclusion

The Smart Automatic Street Light Control System demonstrates practical automation, sensor interfacing, and energy conservation using embedded systems while serving as a foundation for future real-hardware implementation.